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ABSTRACT

A field trip provides an essential opportunity for students to gain practical knowledge by visiting institutions and observing real-world applications of theoretical concepts. On 24 and 25 September, the MSc students of the Department of Statistics and Data Science at the Islamic University, Kushtia, undertook an educational field trip to two renowned national organizations: the **Bangladesh Bureau of Statistics (BBS)** and the **International Centre for Diarrhoeal Disease Research, Bangladesh (ICDDR, B)**.

The visit to BBS offered valuable insights into the national statistical system, including census and survey operations, data collection techniques, digital enumeration processes, and the challenges faced in ensuring data accuracy for evidence-based policymaking. Students also learned about the organizational structure of BBS, and the statistical tools and methodologies used in producing official statistics.

At ICDDR, B, students observed how health research is conducted through advanced laboratory practices, epidemiological surveillance, and clinical trials. The visit enhanced understanding of how statistical models and data analysis contribute to public health research, disease prevention, and policy development.

Overall, the trip enriched our knowledge of applied statistics and demonstrated the critical roles that BBS and ICDDR, B play in national development and public health advancement.

Keywords: BBS, ICDDRB, Statistical Systems, Public Health Research, Data Analysis.

Table of Contents

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Chapter I: Introduction

1.1 Motivation

The value of real-life experience in education cannot be overstated. While images and descriptions can enhance understanding, direct observation provides a level of clarity and depth that theoretical study alone cannot achieve. In academic life, especially in subjects like Statistics, much of our learning is centered on theoretical concepts, formulas, and structured examples. However, real-world applications often deviate from textbook situations, requiring practical adjustments and contextual understanding. Therefore, to bridge the gap between theory and practice, field visits play an essential role by allowing students to witness the implementation of statistical methods firsthand.

Educational field trips have long been recognized as an effective teaching and learning strategy. According to Krepel and Duvall (1981), a field trip is an educational excursion that enables students to interact with real environments, displays, or operational systems to develop experiential connections with academic concepts. Previous studies (Krakowka, 2012; Anderson & Zhang, 2003; Davidson et al., 2010; Kisiel, 2000, 2005) highlight that field trips provide unique learning experiences that cannot be replicated within the classroom, offering exposure to new environments, motivation for lifelong learning, enriched engagement, and practical insight into professional practices.

As students of the **MSc 1st semester**, the Department of Statistics and Data Science at Islamic University, Kushtia arranged an educational field trip as part of the **3-credit course requirement for STAT-5105**. This year, the department selected two prominent national and international institutions—**the Bangladesh Bureau of Statistics (BBS)** and **the International Centre for Diarrhoeal Disease Research, Bangladesh (ICDDR,B)**—for this academic excursion. The goal was to provide us with practical exposure to statistical systems, data processes, and real-world applications of the methods we study in theory.

The **Bangladesh Bureau of Statistics (BBS)**, as the national statistical organization of Bangladesh, conducts censuses, socioeconomic surveys, and the Sample Vital Registration System (SVRS), among many other data programs. Visiting BBS gave us the opportunity to observe how national-level data are collected, processed, and analyzed using modern tools such as advanced sampling techniques, and software-based data processing.

While previous student batches visited institutions like BRRI, BARI, RDA, and BARD, our session chose BBS because of its central role in producing the statistical backbone of national planning and evidence-based policy formulation.

On the other hand, **ICDDR,B** is internationally recognized for its groundbreaking work in public health, epidemiology, infectious disease research, and clinical trials. Since 1978, the institution has contributed significantly to global health—most notably through the development of Oral Rehydration Therapy (ORT). For us as future statisticians, the visit offered a unique chance to observe how statistical methodologies are embedded within epidemiological surveillance systems, laboratory research, biostatistical modeling, and clinical study design. It enriched our understanding of practical biostatistics and health-related data analysis.

Our motivation for selecting BBS and ICDDR,B originated from our aspiration to broaden our academic knowledge, gain hands-on experience, and witness how statistical theories transform into real-world applications. Both institutions utilize advanced methodologies—including stratified sampling, cluster sampling, surveillance sampling, regression modeling, and experimental design—making them ideal learning environments for statistics students.

Thus, the field trip was designed to provide us with direct exposure to how data are gathered, validated, processed, and interpreted in real operational settings. This experience is invaluable for developing our understanding of evidence-based planning, informed policymaking, and impactful scientific research, ultimately preparing us for our future careers in statistical and data-driven professions.

1.2 Transport

We traveled from our campus by a university bus arranged by the department. The bus had sufficient seating capacity for all participants, ensuring a comfortable journey. Although its overall performance was satisfactory, there were minor limitations. However, the driver remained patient, cooperative, and supportive throughout the entire trip, contributing positively to the travel experience.

1.3 Purpose of the Trip

The primary purposes of the field trip were:

1. **To visit the Bangladesh Bureau of Statistics (BBS)** and gain an understanding of its organizational structure, data collection procedures, and the statistical methodologies used in national censuses and surveys.
2. **To visit the International Centre for Diarrhoeal Disease Research, Bangladesh (ICDDR,B)** and learn about the statistical and epidemiological techniques applied in public health research, laboratory investigations, and clinical studies.
3. **To enhance practical knowledge** by observing how real-world institutions employ statistical models, sampling techniques, and data processing systems.

1.4 Organization of the Report

This report is organized into several chapters for clarity and coherence:

- **Chapter 1** presents the introduction, including motivation, objectives, scope, methodology, and overall structure of the report.
- **Chapter 2** provides a detailed overview of BBS, the sessions conducted during our visit, the methodologies discussed, and the key lessons learned.
- **Chapter 3** describes the visit to ICDDR,B, including departmental interactions, research activities observed, methodologies applied, and major findings.
- **Chapter 4** presents a comparative analysis of BBS and ICDDR,B, focusing on differences in statistical practices, data systems, and institutional functions.
- **Chapter 5** demonstrates real-life applications using selected methodologies learned from both institutions.
- **Chapter 6** concludes the report with final remarks, reflections, and suggestions for future improvements or further study.

Chapter 02: Literature Review

(Joarder & Islam, 2019) emphasize that BBS serves as the backbone of Bangladesh's national evidence framework, generating official statistics through censuses, household surveys, and administrative data systems. Their research shows that BBS data are essential for monitoring economic growth, poverty, education, gender equality, and overall progress toward Sustainable Development Goals (SDGs).

(World Bank, 2018) identifies BBS as one of the most important public-sector institutions in Bangladesh, providing high-quality datasets such as the Household Income and Expenditure Survey (HIES), Labour Force Survey (LFS), and Agriculture Census. These datasets support national planning, budget allocation, and economic modelling.

(UNFPA, 2021) documents how BBS has modernized its data collection systems, shifting from paper-based enumeration to Computer-Assisted Personal Interviewing (CAPI). This transition improved data accuracy, reduced error rates, and enabled near real-time monitoring during national surveys like MICS and HIES.

(BBS, 2020) through the Household Production Satellite Account (HPSA), introduces a method for valuing unpaid domestic and care work. This work is crucial for recognizing women's invisible contributions and supporting gender-responsive national accounting—an approach aligned with international best practices recommended by UN Women.

(BBS, 2022) highlights how homestead food production improves dietary diversity, micronutrient intake, and women's nutritional status. This aligns with global evidence showing that women-led homestead agriculture significantly improves food security outcomes (FAO, 2021).

(BBS & UNICEF, 2019) through MICS 2019 provides high-quality district-level estimates on maternal health, child nutrition, education, WASH, immunization, and child protection. MICS serves as a global gold standard survey used in 118 countries to track SDGs—making BBS a key contributor to international development monitoring.

(Global Adult Tobacco Survey, BBS 2017) provides nationally representative estimates on tobacco use, informing national tobacco control laws, taxation policies, and public health campaigns.

(BBS, 2021a) gender-based employment and wage statistics highlight persistent gender inequality in labour force participation, occupational segregation, and wage disparity. These findings align with global literature by the International Labour Organization (ILO, 2020) highlighting similar trends in lower-middle-income countries.

(icddr,b, 2018) describes icddr,b as a global leader in public health research, best known for developing Oral Rehydration Solution (ORS)—recognized as one of the most important medical discoveries of the 20th century, credited with saving over 50 million lives worldwide.

(Cash et al., 2014) document the scientific breakthroughs behind ORT (Oral Rehydration Therapy), illustrating how research conducted at icddr,b shaped global diarrhoeal disease management strategies endorsed by WHO and UNICEF.

(Larson et al., 2019) highlight icddr,b's unique Health and Demographic Surveillance Systems (HDSS) in Matlab, Chakaria, and urban slums—the longest-running surveillance sites in the Global South. These platforms provide continuous, high-quality longitudinal data on births, deaths, fertility, migration, and disease patterns.

(The Lancet, 2018) recognizes Matlab HDSS as a model for community-based health research worldwide, underpinning landmark studies on family planning, immunization, maternal health, and child survival.

(UNICEF, 2020) shows that icddr,b-led maternal and newborn health interventions—such as community-based newborn care and Kangaroo Mother Care—significantly reduce neonatal mortality and improve early childhood outcomes.

(WASH Consortium, 2021) reports that icddr,b's work on water, sanitation, hygiene (WASH), and environmental contamination provides critical insights into diarrhoeal diseases, antimicrobial resistance, and climate-change-related health threats.

Chapter 3: Observations and Institutional Activities at BBS and ICDDR, B

3.1 Bangladesh Bureau of Statistics (BBS)

BBS Building (Parisankhyan Bhaban)

5

Overview

The **Bangladesh Bureau of Statistics (BBS)**, operating under the Statistics and Informatics Division of the Ministry of Planning, serves as the **sole National Statistical Office (NSO)** of Bangladesh. Since its establishment, BBS has been responsible for systematically collecting, compiling, and disseminating official statistics related to population, agriculture, industry, the economy, demography, and numerous socio-economic sectors. The institution continuously adapts to emerging data needs and user demands to ensure the availability of accurate and reliable information.

To remain aligned with modern information practices, BBS publishes its statistical reports both in traditional printed form and as electronic versions through its official website. These resources are accessible to government policymakers, researchers, academicians, development partners, students, and all other users who rely on authentic national statistics for planning, research, and decision-making.

Background and Establishment

Following the Liberation War of 1971, the Government of Bangladesh recognized the need for a unified and organized statistical system to support national planning and development. In August 1974, four separate statistical organizations were merged to form BBS:

- The Statistics Bureau (Ministry of Planning)
- The Agricultural Statistics Bureau (Ministry of Agriculture)
- The Agricultural Commission (Ministry of Agriculture)
- The Census Commission (Ministry of Home Affairs)

Initially, BBS operated without a formal legal framework. This changed with the enactment of the **Statistics Act, 2013**, which provided a legal foundation for statistical activities and strengthened the authority and structure of BBS. The Act formally established BBS through a gazette notification on **3 March 2013**, marking a significant milestone in the country's statistical development.

Session Summary

- We arrived BBS building at 9am. After finishing some preparations we waited at conference room.
- At 9.45am DG of BBS warmly welcomed us. We were very pleasant to meet with him.
- At 9.50am he introduced us with Director, SSTI, BBS. We were very lucky to meet with such an honorable person, his teaching methods were amazing.
- At 10.30am we take a snack break and continue our session.
- At 11.15 am DDG of BBS, Ghose Subobrata, came to meet with us.
- At 1.00pm we had our prayer as well as lunch break. At 2.00pm we resumed our session
- At 2.30pm Joint Director of the Demography and Health Wing, BBS started his session.
- At 3.00pm Deputy Director, Computer Wing, BBS and also guide us and showed us around; the museum, library, press and computer cells.

What We Learn?

After passing a whole day in on BBS and spending times with our honorable techers We learned the following things:

What is BBS ?

BBS is the centralized official bureau in Bangladesh for collecting statistics on demographics, -ics, the economy, and other facts about the country and disseminating the information.

Vision of BBS

To become a nationally and internationally recognized center of excellence in official statistics.

Mission

- To provide accurate, standardized, and timely official statistics;
- To supply data based on the needs of policymakers, planners, researchers, and other users;
- To enhance institutional efficiency and service quality;
- To promote professionalism and international statistical standards.

Functions of the Bangladesh Bureau of Statistics (BBS)

In accordance with the **Statistics Act, 2013**, the Bangladesh Bureau of Statistics (BBS) performs a wide range of functions aimed at producing reliable and timely official statistics. The major functions of BBS include the following:

- Producing, preserving, and disseminating accurate, credible, and timely statistical information.
- Conducting surveys across various socio-economic sectors to generate high-quality statistical data.

- Organizing and implementing all major censuses, including the Population and Housing Census, Agricultural Census, Fisheries and Livestock Census, Economic Census, and other national surveys.
- Providing user-friendly and reliable statistics to government planners, policymakers, researchers, educational institutions, development partners, and other national and international stakeholders.
- Formulating statistical policies, standards, and procedures to ensure consistency and quality across all statistical activities.
- Supervising and monitoring branch, district, and upazila statistical offices, and reviewing reports where necessary.
- Implementing and periodically updating the **National Strategy for the Development of Statistics (NSDS)**.
- Conducting training programs to enhance the skills and capacity of human resources in the statistical sector.
- Promoting public awareness regarding the importance and usefulness of statistics in national development.
- Ensuring the integration and use of modern information and communication technology (ICT) in statistical operations.
- Coordinating and collaborating with national authorities, advisory bodies, NGOs, and international organizations on statistical matters.
- Compiling major economic indicators, including the Consumer Price Index (CPI), other price indices, and national accounts.
- Compiling and disseminating economic, environmental, demographic, health, and social indicators.
- Estimating production and land use statistics for major and minor crops, including cost of production and crop area.
- Developing, maintaining, and updating a unified **Geo-code system** for all government institutions.
- Creating and updating the **National Population Register (NPR)**.
- Developing an integrated central Geographic Information System (GIS).
- Standardizing statistical programs according to international guidelines and best practices.
- Establishing and maintaining a National Data Bank with modern archival systems and alternative storage facilities.

- Certifying the quality and standards of official statistics for national and international users.
- Providing statistical consultancy and advisory services.

National Strategy for the Development of Statistics (NSDS)

The **NSDS** is a national framework designed to modernize and strengthen the statistical system of Bangladesh. Originating from the global Marrakech Action Plan for Statistics (MAPS) in 2004, the NSDS was adopted in Bangladesh and approved by the Cabinet in **October 2013**.

The strategy outlines 61 detailed strategic goals aimed at:

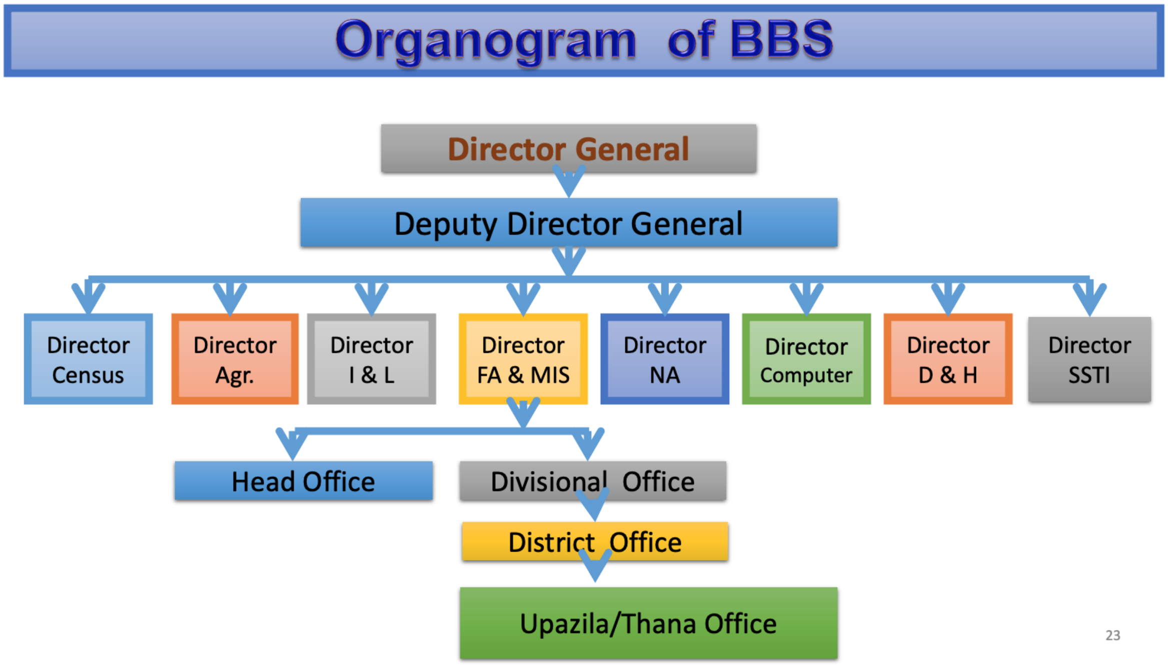
- Modernizing data production and dissemination;
- Standardizing statistical methods;
- Strengthening data quality, reliability, and timeliness;
- Enhancing national capacity for evidence-based planning;
- Improving data storage, networking, and national income measurement systems.

Through NSDS, a strong foundation has been established for sustainable, high-quality national statistics.

Organizational Structure of BBS

The **Bangladesh Bureau of Statistics (BBS)** serves as the central institution within the country's **National Statistical System (NSS)**. To ensure more effective supervision of national statistical activities, the former Statistics Division was reorganized in 2010 and later expanded in 2012, becoming the **Statistics and Informatics Division (SID)** under the Ministry of Planning.

BBS is headed by a **Director General (DG)**, who is supported by a hierarchy of officials including the Deputy Director General, Directors, Joint Directors, Deputy Directors, and other technical and administrative staff. According to its current organogram, BBS has a workforce of approximately **4,352 personnel**.



The organizational operations of BBS are carried out through **eight wings**, classified into **five subject-matter wings** and **three service wings**:

Subject-Matter Wings

1. **Census Wing**
2. **Industry and Labor Wing**
3. **Agriculture Wing**
4. **National Accounting Wing**
5. **Demography and Health Wing**

Service Wings

1. **Computer Wing**
2. **Financial Administration and Management Information Systems (FA&MIS) Wing**
3. **Statistical Staff Training Institute (SSTI)**

Each wing is led by a director and is responsible for specific functions related to statistical production, management, or institutional support. Under the FA&MIS Wing, field-level offices—such as Divisional, District, and Upazila Statistical Offices—operate across the country. These offices collect primary data under the guidance of the Headquarters and Divisional Offices, after which the data are forwarded to the relevant wings and project units for processing and analysis.

Core Surveys Conducted by BBS

Survey	Major Outputs	Latest Round
Household Income and Expenditure Survey (HIES)	Poverty measurement, consumption, income	2022
Quarterly Labour Force Survey (LFS)	Estimation of labour force by sex, unemployment rate, industry and occupation	2024
Survey of Manufacturing Industries (SMI)	Gross output, GVA, Employment,	2019
Agriculture Crop Production Survey (ACPS)	Crop-wise production, yield rate, Agriculture wage by gender	2022 & 2023 Continuous
. Price and Wage Rate Survey (PAWARS)	Computation of CPI, WPI/PPI, wage rates etc.	October 2023 & Continuous
Sample Vital Registration Survey (SVRS)	Population growth rate, birth and death rates, migration etc.	2022 & 2023 Continuous
Health and Morbidity Status Survey (HMSS)	Disability, health expenditure, morbidity, maternal childcare etc.	2014
Multiple Indicator Cluster Survey & Child Nutrition Survey (MICS/CNS)	Primary level school attendance, infant mortality child labour, sanitation child nutrition etc.	2019
Literacy Assessment Survey	Functional Literacy Rate	2023

Public Opinion Survey on National Reform

- Constitutional Reform Survey
- Local Government Reform Survey
- Media Reform Survey
- Electoral System Reform Survey
- Health Sector Reform Survey
- Public Opinion Survey on National Reform

Ongoing Reform-Related Surveys Conducted by BBS

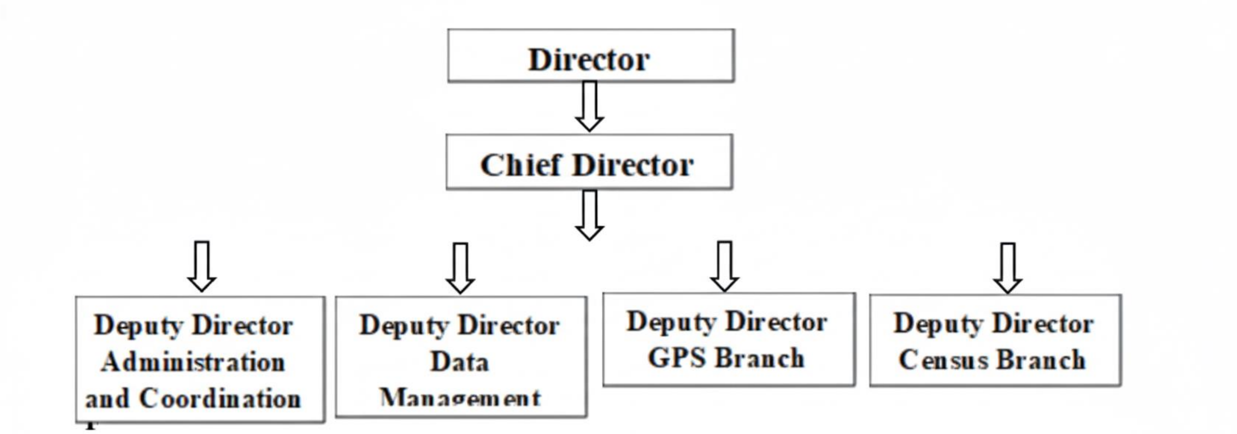
- National Parliamentary Election Survey
- Pay Commission Survey
- Living Standards Survey

3.1.1 Census wings

The **Census Wing** is responsible for organizing and conducting three major decennial censuses in Bangladesh: the **Population and Housing Census**, the **Agricultural Census**, and the **Economic Census**. Following independence, the country carried out its first Population and Housing Census in **1974**, the first Agricultural Census in **1977**, and the first Economic Census in **1986**. Since then, Bangladesh has conducted **six Population and Housing Censuses**, **five Agricultural Censuses**, and **three Economic Censuses**, providing essential data for national planning and development.

In addition to these core responsibilities, the Census Wing undertakes several large-scale periodic surveys, including the **Literacy Attainment Survey (LAS)**, which gathers empirical information on literacy and education levels across the country. Recently, new initiatives such as the **Slum Census** and the **Floating Population Census** have been introduced to address data gaps related to marginalized groups and populations not captured through traditional census operations.

Census Wing – Organizational Structure



Functions of the Census Wing

- Conduct major national censuses, including the Population and Housing Census, periodic specialized censuses, and education- and labour-related surveys.
- Implement various censuses and surveys based on the requirements and directives of the Government of Bangladesh.
- Produce intercensal population estimates and prepare mid-term population projections using demographic indicators between two census periods.
- In accordance with Section 11 of the **Statistics Act, 2013**, and following the *Guidelines for Preparing and Publishing Statistics* (2014 and 2016), review and issue *Non-Objection Certificates (NOC)* for censuses and surveys carried out by government agencies other than the designated authority.
- Provide technical assistance to national and international organizations in designing sampling frameworks for different surveys.
- Assist ministries and agencies in managing **68 national indicators** related to crime, governance, security, socio-economic conditions, and education.
- Offer continuous support to working groups under the **National Indicator Coordination Committee**, particularly those responsible for crime, governance, security, socio-economic, and education-related data activities.

Functions of the Administration & Coordination Section

- Manage administrative, financial, and coordination-related activities of the Wing.
- Handle employee-related matters, including leave, retirement, PRL, pension, and family pension processing.
- Issue various official certificates for officers and employees, including non-marriage certificates.
- Maintain and manage logistics required for the Wing and take necessary initiatives for logistical support.
- Prepare, compile, and dispatch various official reports.
- Perform additional responsibilities assigned by the Joint Director or Director.

Functions of the Data Management Section

- When statistics are not produced directly by the Bureau, ensure that relevant ministries, divisions, and subordinate agencies follow BBS guidelines and methodologies while preparing statistical data, and provide necessary approval.
- Provide technical assistance to ministries, divisions, and agencies on survey methodology, sampling techniques, and other statistical issues.
- Supply population-related, education-related, and other required data to different government bodies.
- Approve statistics prepared by relevant institutions, offer technical support during implementation, and monitor data collection activities.
- Support the Secondary and Higher Education Division in implementing the **Capacity Development for Education (CapEd) Phase II** in collaboration with UNESCO.

Ongoing Activities of the Data Management Section

- Assist in collecting and safeguarding data from various ministries and departments, including contributions to SDG-related initiatives and activities under the National Coordination Program for crime, justice, economy, and security.
- Conduct surveys related to security and social issues, such as Household Assessment Surveys and Education Household Surveys.

- Implement data management activities aligned with **SDG Indicator 9.1.1**.
- Periodically provide data to different units within the Bureau, including AIR and IRD cells.
- Carry out data collection, processing, and analysis for social surveys.
- Coordinate nationwide censuses and surveys in collaboration with central and field-level offices.
- Support additional tasks assigned by the Joint Director and Director.

Activities of the Census Wing

- Develops and implements both regular and pilot projects for conducting the Population and Housing Census.
- Conducts follow-up activities for completed censuses in accordance with the Census Wing's operational calendar.
- Plans and designs future census programs and projects.
- Maintains and manages national, divisional, and operational statistical infrastructure to support census implementation.
- Provides technical assistance to registered census units throughout the census process.
- Performs all activities related to census preparation, execution, and documentation.
- Supports relevant ministries and departments in implementing census operations when required.
- Prepares, compiles, and disseminates statistical reports and related documents.
- Assists with additional tasks assigned by the Joint Director and Director.

Activities of the Field Report Wing

- Collects perception-based data through field-level surveys and assessments.
- Implements pilot and regular activities related to **SDG Indicator 9.1.2** under the Bangladesh component.
- Conducts cultural and social surveys at the national level.
- Executes various field report–related activities as required.

- Provides guidance and support to ministries and agencies involved in field data collection activities.
- Plays an essential role in implementing national activities of the Wing under the **National Strategy for the Development of Statistics (NSDS)**.
- Supports ongoing operations and any additional responsibilities assigned by the Joint Director and Director.

National Population Register (NPR)

The **National Population Register (NPR)** is a centralized database designed to systematically store both statistical and individual-level information on all citizens and residents of Bangladesh. It is intended to support administrative functions by providing accurate data for planning, identification, and service delivery.

Citizens and residents provide updated information—such as citizenship status and personal details—every six months, ensuring that the database remains current and reliable.

Objectives of the NPR

- To develop a comprehensive and integrated national database.
- To ensure accurate identification of individuals for administrative and governance purposes.
- To support the implementation of government programs and enhance service delivery.
- To establish a unified repository containing citizen-related information.
- To centralize data management for accurate population registration.

Potential Outputs of the Bangladesh NPR

- Support effective implementation of government security and administrative services.
- Generate a complete household- and individual-level statistical database.
- Collect and maintain geographic and boundary information for electoral areas.
- Provide population-related data essential for public service delivery.
- Produce national and international statistical reports.

- Facilitate population registration through the National Register.
- Establish sampling frames for national surveys.
- Systematically compile administrative and socio-economic data.
- Produce statistics on vital events, health, education, employment, income, and other essential sectors.

3.1.2 Industry and Labor Wing

Bangladesh is increasingly recognized as one of the world’s fast-growing economies, and the industrial sector plays a crucial role in sustaining this growth. The **Industry and Labor Wing** of BBS is responsible for compiling, analyzing, and publishing statistics related to industrial structure, productivity, technological advancement, employment trends, and other key macroeconomic indicators. These data support national economic planning and provide insights into the performance of various industrial sectors.

Labor statistics refer to the systematic collection, analysis, and dissemination of data on employment and the labor market. Findings from labor force surveys offer essential information about labor market efficiency, job characteristics, demographic patterns, and shifts in employment. Such information enables government agencies, policymakers, businesses, researchers, and the general public to make informed decisions. Labor statistics are also vital for ensuring worker rights, identifying labor market risks, and shaping policies that improve working conditions, expand employment opportunities, and promote inclusive economic growth.

Internationally, labor force surveys have a long history. The United States first introduced a monthly survey in 1940, followed by Canada in 1945 (quarterly) and 1952 (monthly). France initiated its labor force survey in 1950 and has maintained regular assessments since 1960. Germany and Sweden began their surveys in 1957, Australia in 1960, and the United Kingdom in 1973. These longstanding initiatives highlight the global importance of labor statistics in understanding employment dynamics over time.

Key Indicators Measured in the Labor Force Survey

- Unemployment
- Employment by Industry
- Employment by Occupation
- Employment by Job Status
- Employment in the Informal Economy
- Employment–Population Ratio
- Youth Unemployment
- Labour Underutilization
- Labor Force Participation Rate
- Disabled Population Rate in the Labor Force
- Hours of Work
- Monthly Wage
- Part-Time Employment
- Migration Costs

Survey Methodology

All individuals aged **15 years and above** included in the survey are classified into three categories:

1. **Employed**,
2. **Unemployed**, and
3. **Outside the Labor Force**.

For analytical purposes, individuals aged **15–29 years** are considered **youth**.

The questionnaire used in the **2022 Labor Force Survey (LFS)** was aligned with the standards of the **19th ICLS Resolution**, ensuring international comparability of data. For the first time, the survey collected employment information on **persons with disabilities**, aiding the development of inclusive labor market policies. Additionally, a **migration module** was introduced to capture migration-related data.

The 2022 survey also marked the first use of **CAPI (Computer Assisted Personal Interviewing)**, enhancing accuracy and efficiency in data collection.

Coding Method

To classify economic activities, the survey used the **Bangladesh Standard Industrial Classification (BSIC) 2020**, which is based on the **International Standard Industrial Classification (ISIC Rev. 4)**. BSIC 2020 groups all economic activities into **21 sections**, and statistics are presented under three broad sectors:

- **Agriculture**,
- **Industry**, and
- **Services**.

Similarly, for occupational classification, the survey followed the **Bangladesh Standard Classification of Occupations (BSCO) 2020**, modeled after the **International Standard Classification of Occupations (ISCO 2008)** of the International Labour Organization (ILO). Under BSCO, occupations are divided into **10 major groups**:

1. Managers
2. Professionals
3. Technicians and Associate Professionals
4. Clerical Support Workers
5. Service and Sales Workers
6. Skilled Agricultural, Forestry, and Fishery Workers
7. Craft and Related Trades Workers
8. Plant and Machine Operators and Assemblers
9. Elementary Occupations
10. Other Occupations

Definitions in progress

Employment

Employed: According to the ILO guideline, persons who have worked for at least one hour during the last 7 days in exchange for wages, salary, or profit, or have engaged in production activities for own household consumption

Own use production work

Production for household's own consumption: Any household member aged 15 years or above who has spent at least one hour during the last week in producing goods and services for their own family's consumption—such as agricultural activities and fish farming/fishing (e.g., producing rice, wheat, potatoes, vegetables, etc., or rearing livestock)—is considered engaged in production for household's own consumption.

Not in labour force

Out of the labour force: Persons who are generally considered outside the labour force include students, the sick, the elderly, those unable to work, retirees, and homemakers who did not do any work during the last seven days, did not seek any work in the last 30 days, and were not available for work in the last seven days.

Unemployment

Unemployed: A person aged 15 years or above will be considered unemployed if he or she has sought any kind of work for wages, salary, profit, or own-product production during the last 30 days and was available for work during the last 7 days.

3.1.3 Agriculture Wing

The **Agriculture Wing** is responsible for producing and disseminating comprehensive statistical data related to the agricultural sector of Bangladesh. Its activities support national planning, food security assessments, and agricultural policy formulation.

Key Functions

- Prepare and publish annual estimates of land area, production volume, and yield rates for six major crops—**Aus, Aman, Boro, Wheat, Jute, and Potato**—as well as approximately **140 minor crops**, including vegetables, fruits, and flowers.
- Generate seasonal forecasts on cultivated land and projected crop production to assist in agricultural planning.
- Estimate crop losses caused by natural disasters, assess agricultural labour wages by gender, and compile land use and irrigation statistics. These findings are published in the **Yearbook of Agricultural Statistics**.
- Compile and consolidate forest and fisheries statistics using data provided by the respective departments.
- Prepare all agricultural statistics in accordance with international standards and guidelines set by the **Food and Agriculture Organization (FAO)**.

3.1.4 National Accounting Wing

The **National Accounting Wing** is responsible for collecting, compiling, and disseminating the national accounts statistics of Bangladesh. Its core functions include the preparation of key macroeconomic indicators such as Gross Domestic Product (GDP) and other aggregates, including Gross National Income (GNI), investment and savings estimates, private consumption, government expenditure, per capita income, and GDP growth rates. The wing also generates various price and wage statistics, including the Consumer Price Index (CPI), Wage Rate Index (WRI), Building Material Price Index (BMPI), and House Rent Index (HRI). In addition, it compiles the Quantum Index of Industrial Production (QIIP), the Producer Price Index (PPI), and the Foreign Trade Statistics (FTS). The **Household Income and Expenditure Survey (HIES)**, an essential source for poverty estimation and socioeconomic analysis, is also administered under this wing.

Vision & Mission of the Wing

Vision:

- Empower Bangladesh's National Accounts system to meet global standards.

Mission:

- Compile and provide demand-based National Accounts Statistics (NAS) in accordance with the Bangladesh Statistics Act.
- Follow official statistical principles in all statistical activities.
- Apply international standards of statistical classification following the System of National Accounts (SNA).

Current Activities of the Wing:

The National Accounting Wing consists of the following **9 branches**:

1. National Income Branch (Services, Industry, and Agriculture)
2. National Expenditure Branch
3. Foreign Trade Branch
4. Price and Wage Statistics Branch
5. Current Production Statistics Branch
6. Manuscript (Publication) Branch
7. Administration Branch
8. Quarterly GDP Projection Branch
9. Economic Statistics Projection Branch

Branch-Based Activities

National Income Branch

Responsibilities:

- Estimate Gross Value Added (GVA) for 19 sectors using the production approach following SNA and ISIC.
- Compile Gross Domestic Product (GDP) and Gross National Income (GNI) by aggregating sector-wise value added.
- Calculate Per Capita GNI.
- Perform tasks related to changing the GDP base year.
- Provide national accounts data to UNSD, ADB, World Bank, Bangladesh Bank, etc.
- Contribute regularly to the Ministry of Finance's *Bangladesh Economic Survey*.
- Ensure accurate data presentation for government planning.
- Prepare, publish, and document national accounts statistics.

National Expenditure Branch

Responsibilities:

- Prepare expenditure-based national accounts as per SNA standards.
- Compile estimates of consumption, investment, and savings.
- Collect export and import data to determine net factor income and net current transfers.
- Collect, classify, and process budgets from City Corporations, District Councils, Municipalities, Union Councils, and autonomous institutions.
- Collect, classify, and process revenue and development (ADP) budgets.

Foreign Trade Branch

Responsibilities:

- Collect monthly import/export data from the National Board of Revenue (NBR).
- Publish *Foreign Trade Statistics of Bangladesh*.
- Issue the monthly *Foreign Trade Statistics (FTS)* report.
- Compile the unit price index for exports and imports.
- Prepare the unit value index.

Price and Wage Statistics Branch

Responsibilities:

- Compile major price and wage statistics such as:
    - Consumer Price Index (CPI)
    - Inflation Rate
    - Business Price Index (BPPI)
    - Wage Rate Index (WRI)
    - House Rent Index (HRI)
- Currently uses **2021–22** as the base year for price and wage statistics.
- Collect monthly price data from **1,545 markets** across the country:
    - One rural market per district
    - One urban market per district
    - 12 markets/outlets in Dhaka city

International Activities

Istanbul Programme of Action (IPoA)

Responsibilities:

- Provide estimates for Bangladesh's three criteria for graduation from the LDC category:
    - **Human Assets Index (HAI)**
    - **Economic Vulnerability Index (EVI)**
    - **Gross National Income (GNI)**
- Prepare GNI using the **Atlas Method**.
- Compile the composite index for **HAI** using six indicators.
- Compile the composite index for **EVI** using eight indicators.

Future Planning

- Compilation of **Regional (District) Accounts**
- Compilation of **Supply and Use Tables (SUT)**
- Compilation of **Input–Output Tables (IOT)**
- Compilation of **Institutional Sector Accounts (ISA)**
- Development of an **Index of Service Production (ISP)**
- Experimental compilation of **Blue Economy** contribution (portion of GDP)

Limitations and Challenges

- Need for new surveys and case studies to reduce **undercoverage**.
- Regular **base year updating and rebasing** of GDP every 10 years.
- Regular base year changes for all indices.
- Methodological development requirements.
- Need for improved **resource mobilization**.
- Institutionalization of technical expertise.
- Capacity building and training gaps.

3.1.5 Demography and Health Wing

The **Demography and Health Wing** is responsible for collecting, compiling, analyzing, and disseminating statistics related to health, demography, nutrition, and various socio-economic indicators.

Functions of the Demography and Health Wing

- Collect, compile, analyze, and disseminate demographic, health, nutrition, and socio-economic data.
- Conduct intercensal demographic surveys on vital events such as fertility, mortality, life expectancy, disability, and nuptiality.
- Carry out surveys aimed at generating data on health conditions, morbidity, and nutritional status.
- Produce, compile, and disseminate **gender statistics**.

Sections of the Wing

1. Admin, Finance & Coordination Section

- Manage all administrative activities of the wing.
- Ensure staff attendance and maintain discipline.
- Coordinate the functions of different sections within the wing.
- Organize seminars, workshops, and training programs.
- Prepare periodic reports for the Cabinet Division, SID, and other organizations.

2. Demographic Statistics Section

- Conduct demographic surveys and carry out in-depth analysis.
- Provide demographic data to users and stakeholders based on demand.

Health & Nutrition Statistics Section

- Conduct surveys related to health and nutrition and perform in-depth analytical work.
- Supply health and nutrition statistics to different users and stakeholders.

Socio-Economic Statistics Section

- Conduct socio-economic surveys and perform detailed analysis.
- Provide socio-economic data according to user and stakeholder requirements.

Gender Statistics Cell

To enhance the development of gender-focused statistics, BBS established a **Gender Statistics Cell** under this wing on **07 February 2017**.

Major Activities:

- Produce, compile, and disseminate gender statistics.
- Coordinate with relevant organizations, development partners, and researchers, and organize workshops, seminars, and meetings for the improvement of gender statistics.
- Ensure gender-disaggregated data in all censuses and surveys conducted by BBS.

Ongoing Activities

- Monitoring the **Situation of Vital Statistics of Bangladesh**.
- Conducting the **Violence Against Women (VAW) 2025** survey.
- Preparing the **Satellite Account on Unpaid Domestic and Care Work** (completed).
- Conducting the **Multiple Indicator Cluster Survey (MICS) 2025**.
- Implementing the **Health and Morbidity Status Survey 2024** under the *Improving Demographic and Health Statistics of Bangladesh Project*.

Planned Activities

- Qualitative Survey on **Violence Against Women (VAW) 2025**
- **National Survey on the Elderly Population in Bangladesh 2025**
- **Survey on Accidental Injury in Bangladesh 2026**
- Compilation of **Health Statistics**
- Compilation of **Gender Statistics 2025**

3.1.6 Computer Wing

The **Computer Wing** provides comprehensive IT support to all subject-matter wings of BBS.

Key Functions

- Data storage, processing, and tabulation for all censuses and surveys.
- Provide ICT training for BBS officers and staff.
- Plan, install, operate, and maintain servers, databases, and computer networks.
- Manage the official BBS website and ensure electronic data delivery to users.

3.1.7 FA & MIS Wing

The **Finance Administration & Management Information Systems (FA & MIS) Wing** was established after internal reorganization in July 1999.

Branches of the FA & MIS Wing

- Administration, Budget, and Financial Management Branch
- BINIK
- Coordination Branch
- Publication and Printing Branch
- Library

Main Functions

- Restructuring and updating the personnel framework.
- Formulating and revising recruitment rules for officers and employees.
- Managing recruitment, promotion, and retirement approval processes.
- Overseeing administration, budgeting, and financial management across headquarters and field offices.
- Managing audit and legal affairs.
- Maintenance of the Statistics Building, vehicles, and library facilities.
- Printing and publication of all BBS statistical reports.

3.1.8 Statistical Staff Training Institute (SSTI)

Established in **1992**, the **Statistical Staff Training Institute (SSTI)** is responsible for providing professional training to all levels of BBS officers and employees.

Key Functions

- Conduct training in statistics, administration, IT, finance, budgeting, project planning, labor laws, and related fields.
- Implement annual training programs beginning each July to create a skilled and competent workforce.
- Ensure a minimum of **60 hours of training** for all categories of officers.
- Plan for the establishment of a fully developed training academy in the future.

3.2 International Centre for Diarrhoeal Disease Research, Bangladesh (ICDDR,B)

Overview

The International Centre for Diarrhoeal Disease Research, Bangladesh (ICDDR,B), located in Dhaka, is a renowned international health research organization dedicated to saving lives through scientific innovation, treatment, and capacity building. The centre focuses on major global health challenges, including improving neonatal survival, reducing infectious disease burden, and addressing emerging threats such as HIV/AIDS.

ICDDR,B works closely with academic and research institutions across the world. Its activities include advanced research, training, and program-based initiatives that generate and disseminate life-saving knowledge.

It is recognized as one of the most influential research institutes in the Global South, contributing nearly **18% of Bangladesh's scientific publications**, according to the Thomson Reuters Web of Science database.

The organization employs a diverse workforce consisting of local and international professionals—public health scientists, clinicians, laboratory specialists, epidemiologists, nutritionists, demographers, social and behavioral scientists, IT experts, and specialists in infectious diseases and vaccine sciences.

ICDDR,B is supported by around **55 donor countries and organizations**, including Sweden (SIDA), Canada, the UK, Bangladesh, the USA, UN agencies, philanthropic foundations, universities, research institutes, and private sector partners. The centre is overseen by an international Board of Trustees with 17 members from various countries.

History of ICDDR,B

ICDDR,B originated from the **SEATO Cholera Research Laboratory**, established in 1960. After Bangladesh's independence in 1971, its operations reduced temporarily due to funding shortages. A bilateral agreement between the Government of Bangladesh and USAID later restored direct financial support.

By 1978, the centre had achieved significant scientific breakthroughs, particularly in:

- Oral Rehydration Solution (ORS)
- Pathophysiology of shigellosis
- Rotavirus research
- Family planning studies

That same year, leading global scientists recommended elevating the centre to an international research institution. It was formally established as ICDDR,B through a Presidential Ordinance issued by President Ziaur Rahman, and later approved by the Bangladesh Parliament in 1979.

Among ICDDR,B's greatest contributions is the development and global implementation of **Oral Rehydration Therapy (ORT)**—a simple, scalable treatment estimated to have saved **over 50 million lives worldwide**.

Since 1978, the centre has trained more than **27,000 health professionals from over 78 countries** in fields such as diarrhoeal disease management, epidemiology, biostatistics, demographic surveillance, and child survival strategies. As disease-related child mortality has declined, ICDDR,B has increasingly focused on injury prevention, particularly drowning.

Facilities

ICDDR,B's headquarters in **Mohakhali, Dhaka** houses modern laboratories, specialized hospitals, and training centres.

Dhaka Hospital, established in 1962, was originally designed for diarrhoeal disease management. It now treats more than **140,000 patients annually** and serves as a platform for extensive clinical research. Innovations such as a fully paperless patient record system—also adopted at the Matlab field site—were pioneered here.

Matlab Hospital, located about 50 km southeast of Dhaka within the Health and Demographic Surveillance Site, is a **120-bed facility** providing free care, including diarrhoeal disease treatment and maternal/child healthcare, to more than **30,000 patients each year**. It played a significant role in the community trials that informed ORS development and early cholera vaccine research.

Awards and Recognition

ICDDR,B has received prestigious global awards for its pioneering health research and humanitarian impact:

- **2017 – Conrad N. Hilton Humanitarian Prize** (USD 2 million), recognizing contributions to improving health among impoverished populations.

- **2016 – Commendation by UN Secretary-General Ban Ki-moon**, highlighting ICDDR,B's role in accelerating progress in reducing maternal, infant, and child mortality.
- **2001 – Bill & Melinda Gates Foundation's First Gates Award for Global Health**.
- **2002 – Pollin Prize for Pediatric Research**, awarded to scientists involved in ORT development.
- **2006 – Prince Mahidol Award (Public Health)** for contributions to ORT.
- **2007 – Leadership Award from the Alliance for the Prudent Use of Antibiotics**.

These recognitions reflect ICDDR,B's longstanding leadership in global health research, innovation, and humanitarian service.

ICDDR,B Research Framework

1. Overview of ICDDR,B's Research

ICDDR,B conducts cutting-edge research targeting some of the world's most critical public health challenges, with a particular focus on addressing unmet needs in low- and middle-income countries. In Bangladesh, the organisation benefits from a strong research ecosystem supported by:

- **Advanced laboratory facilities**
- **A major tertiary-level hospital and multiple clinical care units**
- **Well-established field sites** that enable large-scale population-based studies

ICDDR,B collaborates extensively with research institutions and implementation partners across both the global North and South. These partnerships enhance knowledge exchange and help develop solutions that are contextually appropriate and globally relevant.

2. ICDDR,B Research Goals

✓ Addressing Climate-Related Health Risks

Conduct research on the health impacts of climate change, with special emphasis on vulnerable populations and high-risk communities.

✓ Gender, Sexual, and Reproductive Health and Rights (SRHR)

- Investigate patterns of violence against women and children.
- Design, implement, and evaluate interventions aimed at prevention and response.
- Address child marriage and strengthen SRHR by identifying key drivers and developing evidence-based solutions.

✓ Preventing and Controlling Non-Communicable Diseases (NCDs)

- Examine the growing burden of NCDs—including mental health—in Bangladesh and other LMICs.
- Identify major risk factors and test innovative strategies for prevention and treatment.
- Improve early detection and management of NCDs, especially in underserved and hard-to-reach areas.

✓ Improving Urban Healthcare and Advancing Universal Health Coverage (UHC)

- Identify gaps in urban health systems and propose solutions.
- Promote digital health innovations for patient care, data systems, and decision-support tools.
- Conduct policy research to support progress toward universal health coverage.

✓ Preventing and Controlling Infectious Diseases

- Investigate infectious disease transmission dynamics and associated risk factors.
- Study causes of mortality and morbidity linked to infections in humans and animals.
- Address antimicrobial resistance, microbiome-related health issues, and environmental contamination.
- Evaluate WASH interventions and research vaccines, therapeutics, diagnostics, and epidemic/pandemic preparedness.
- Respond rapidly to public health emergencies and disasters.

✓ Improving Maternal, Newborn, Child, and Adolescent Health

- Identify causes of newborn mortality and improve early childhood development outcomes.
- Enhance early detection and management of high-risk pregnancies.
- Examine risk factors for major health conditions—including mental health—affecting women, children, and adolescents.
- Develop and test innovative health interventions.

✓ Reducing Maternal, Adolescent, and Child Malnutrition

- Study biological and socio-environmental factors contributing to malnutrition.
- Develop and evaluate new interventions to prevent and treat maternal and childhood undernutrition.
- Assess the feasibility, scalability, and impact of nutrition-based innovations.

✓ Enhancing Laboratory Services and Genomics Research

- Expand diagnostic and genomic capabilities.
- Improve access to laboratory and genomics services for patients, researchers, and partner organisations.

✓ Strengthening Hospital Services and Clinical Research

- Enhance the quality of care in ICDDR,B-managed hospitals.
- Foster clinical research that improves treatment protocols and patient outcomes.
- Ensure integration of research findings into clinical practice.

ICDDR, B Research Divisions

ICDDR,B structures its scientific activities across four major research divisions, each dedicated to advancing public health through evidence generation, innovation, and implementation in Bangladesh and beyond:

1. **Health Systems and Population Studies Division (HSPSD)**
2. **Infectious Diseases Division (IDD)**
3. **Nutrition Research Division (NRD)**
4. **Maternal and Child Health Division (MCHD)**

1. Health Systems and Population Studies Division (HSPSD)

HSPSD aims to ensure equitable access to affordable, high-quality, and responsive healthcare for all individuals, regardless of socioeconomic status. The division's work is aligned with the **WHO's six health system building blocks**: **service delivery, health workforce, information systems, medical products, financing, and governance.**

Key Areas of Focus

✓ Climate Change, Health, and Population Science

Researches the health impacts of climate change and develops models to strengthen community resilience and adaptation.

✓ Health Economics and Financing

Explores strategies to reduce the financial burden of healthcare and promote equity in health financing.

✓ Universal Health Coverage & Urban Health

Conducts policy research and community-based interventions aimed at achieving **Universal Health Coverage (UHC)**, with a strong focus on improving access to urban healthcare services and reducing out-of-pocket expenditures.

✓ Programme for HIV and AIDS

Leads epidemiological studies and develops evidence-based, community-focused interventions for key populations at risk of HIV.

✓ Matlab Health Research Centre (MHRC)

A cornerstone of global public health research, MHRC includes Matlab Hospital, which provides free clinical care to **over 80,000 patients annually** while supporting major evaluation studies on health interventions.

HSPSD also operates **three Health and Demographic Surveillance Systems (HDSS)**:

- **Matlab HDSS**
- **Chakaria HDSS**
- **Urban HDSS**

Human Resources Capacity

HSPSD consists of:

- **42 Principal Investigators (PIs)**
- **~625 research and support staff**

21 PhD holders in disciplines including sociology, epidemiology, public health, nutritional epidemiology, economics, statistics, and demography

- Over **45 junior and mid-level researchers** with advanced training
- **11 staff currently pursuing PhDs or Post-Doctoral training** abroad

2. Infectious Diseases Division (IDD)

IDD focuses on major public health threats arising from **antimicrobial resistance**, **enteric and respiratory infections**, and **emerging or re-emerging pathogens**. Research is conducted in collaboration with global partners in the USA, Canada, UK, Japan, Australia, and Europe.

IDD conducts cutting-edge laboratory, clinical, and field research on diseases such as **tuberculosis, cholera, typhoid, SARS-CoV-2**, and **Nipah virus**, while also supporting technological transfer for vaccine production in Bangladesh.

Key Areas of Focus

✓ Programme for Enteric Infections and Vaccines

Investigates disease burden, transmission, and pathogenesis of enteric infections, while developing improved diagnostics, vaccines, and scalable interventions.

✓ Programme for Respiratory Infections

Conducts research and surveillance on respiratory pathogens—including influenza and COVID-19—focusing on transmission dynamics, disease severity, and prevention.

✓ Programme for Emerging and Re-emerging Infections

Works with institutions such as the **IEDCR** to detect, characterize, and respond to emerging infectious disease threats.

✓ Genome Centre (icddr,b Genomics Centre - iGC)

Supports large-scale genomic research and investigates gene–environment interactions. iGC aims to position ICDDR,B as a genomic research leader within the Global South.

3. Nutrition Research Division (NRD)

NRD addresses critical issues related to **maternal and child malnutrition**, **food safety**, **immunobiology**, and **non-communicable diseases**. Research ranges from clinical trials and microbiome studies to evaluation of community-based nutrition programs.

Key Areas of Focus

✓ Maternal and Child Nutrition

Develops and evaluates interventions such as **ready-to-use therapeutic foods (RUTFs)** and **microbiota-directed foods** to prevent and treat malnutrition.

✓ Clinical Research

Conducts hospital-based studies on diarrheal diseases, severe malnutrition, AMR, and neonatal infections. Also investigates links between malnutrition, cognitive development, and enteric diseases.

✓ Food Safety Laboratory

Researches foodborne pathogens and antimicrobial resistance, offering **ISO-compliant diagnostic and testing services** to enhance national food safety.

✓ Immunobiology, Nutrition, and Toxicology

Studies interactions among nutrition, immune function, and toxic exposures, including micro- and macronutrient analyses.

✓ Non-Communicable Diseases (NCDs)

Investigates the burden and causes of NCDs such as diabetes, cardiovascular disease, and cancer, and identifies feasible solutions for prevention and management.

Maternal and Child Health Division (MCHD)

MCHD develops and evaluates interventions to improve the **coverage, quality, and equity** of health services for women, newborns, children, and adolescents. Research spans epidemiology, social science, implementation research, and clinical studies, using a **life-course approach**.

MCHD contributes to national policy formulation and adapts lessons learned to other LMIC settings.

MCHD Structure

The division consists of:

- **Maternal, Newborn and Child Health Programme**
- **Child Development Unit**
- **Gender Equity and Rights Unit**

Key Areas of Work

✓ Maternal, Newborn and Child Health Programme

- Identifies causes of neonatal mortality
- Improves detection and management of high-risk pregnancies
- Develops innovations for maternal, child, and adolescent health
- Investigates mental health and other major health conditions

✓ *Child Development Unit*

- Generates evidence on child development and early childhood interventions
- Studies the impact of malnutrition, toxins, illness, and deprivation
- Designs and evaluates cost-effective, scalable interventions
- Conducts long-term follow-up studies to assess developmental outcomes

✓ *Gender Equity and Rights Unit*

- Investigates links between gender and health
- Conducts research on violence against women and children
- Studies child marriage and SRHR among populations with unique vulnerability
- Develops and evaluates evidence-informed interventions

4. Research Platforms

ICDDR,B operates several world-class research platforms that enable high-quality scientific inquiry, innovation, and implementation research. These platforms include:

- **Clinical Facilities**
- **Research Laboratories**
- **Health and Demographic Surveillance Systems (HDSS)**
- **Matlab Health Research Centre (MHRC)**

A. Clinical Facilities

ICDDR,B's clinical facilities serve as critical platforms for clinical research while providing essential healthcare services to surrounding communities.

Dhaka Hospital

Established in 1962 to address acute diarrhoeal diseases—particularly among children—Dhaka Hospital has evolved into one of the most important clinical centres in Bangladesh. It:

- Treats **more than 140,000 patients annually**, most of them infants and children
- Plays a vital role in **disease surveillance**, **antimicrobial resistance monitoring**, and **clinical training**
- Provides care primarily for **diarrhoeal and respiratory illnesses**
- Saves an estimated **40,000 lives each year** through evidence-based treatment
- Pioneered multiple innovations, including a **paperless patient record system**, now also used at Matlab Hospital

Dhaka Hospital is widely regarded as a model of what can be achieved in a resource-limited setting.

Matlab Hospital

The **120-bed Matlab Hospital**, located within the Matlab Health and Demographic Surveillance Site (HDSS) about 50 km south of Dhaka, provides:

- Free care for diarrhoeal diseases
- Maternal and child health services
- Treatment for **over 30,000 people annually**

Matlab Hospital played a pivotal role in:

- Community trials of **oral rehydration solution (ORS)**
- Early cholera vaccine studies

Recently, it has emphasized **integrated maternal and child healthcare** and the implementation of **evidence-based maternal and neonatal interventions**.

B. Research Laboratories

ICDDR,B’s advanced laboratories support cutting-edge science across infectious diseases, genomics, immunology, nutrition, and environmental health. These facilities are equipped to handle high-level diagnostic, molecular, and biomedical research.

Major laboratory units include:

- **Emerging Infections & Parasitology Laboratory**
- **Genome Centre**
- **Gut–Brain Axis Laboratory**
- **One Health Laboratory**
- **Molecular Ecology and Metagenomics Laboratory**
- **Mucosal Immunology and Vaccinology Unit**
- **Mycobacteriology Laboratory**
- **Virology Laboratory**
- **Animal Resources Facility**

C. Health and Demographic Surveillance Systems (HDSS)

ICDDR,B operates **three HDSS sites**—Matlab, Chakaria, and Urban Slums—each designed to generate longitudinal data critical for understanding population health dynamics.

Field workers visit households every three months to collect data on:

- Births
- Deaths
- Pregnancies
- Migration
- Household changes

HDSS data allow researchers to analyze how social, environmental, demographic, and programmatic factors shape health outcomes. They are essential for monitoring **SDGs**, **climate change impacts**, **migration patterns**, and progress toward **universal health coverage**.

1. Matlab HDSS

- Established in **1966**
- Covers **246,893 individuals** in **60,400 households** across **142 villages**
- The **longest-running HDSS in the Global South**
- Globally recognized as a model for population-based surveillance

- Subdivided into seven blocks; ICDDR,B provides community health services in four, while the government provides services in the remaining three

2. Chakaria HDSS

- Established in **1999**, located **348 km southeast of Dhaka**
- Covers **86,667 individuals** in **16,000 households** across **49 villages**
- Includes coastal, plain, and hilly terrain
- A hotspot for studying **climate change impacts on health**
- Transitioned from paper-based to web-based data collection in **2014**

3. Urban Slum HDSS

- Established in **2015** in Dhaka North, Dhaka South, and Gazipur City Corporations
- Covers **120,000 individuals** in **32,000 households**
- Populations face extreme heat, overcrowding, and poor living conditions—ideal for studying **urban vulnerability and heat island effects**

Combined Coverage

Together, the three HDSS sites cover:

- **453,560 individuals**
- **106,400 households**

These diverse populations provide ideal platforms for monitoring national and global public health goals and evaluating government and NGO health interventions.

D. Matlab: Impact on Public Health

Matlab is ICDDR,B's most influential rural research site and a major global public health resource.

Over **five decades**, Matlab has generated unparalleled population-based data, influencing health policy and practice both nationally and internationally.

The Matlab HDSS

Every two months, field staff document:

- Births and deaths
- Marriages and divorces
- Migrations
- Household changes

Socioeconomic surveys further collect data on:

- Occupations
- Household assets

Research Contributions

Matlab's research has shaped major global health advances:

✓*Family Planning*

Community health worker programmes in the 1970s greatly increased contraceptive use and reduced fertility—models later adopted worldwide.

✓***Immunization***

Research showed that **63% of childhood deaths** could be prevented through vaccination, catalyzing national immunization efforts.

✓***Child Health & Family Planning***

Child mortality dropped by **75% over 25 years** due to integrated programmes.

✓***Longevity***

Life expectancy increased from **50 to 65 years** over the past four decades due to reduced child mortality and lower fertility.

Matlab remains a global model for demographic surveillance and community-based health intervention research.

Collaboration

ICDDR,B prioritizes strong partnerships with national and international collaborators across the Global North and South.

A. Scientific Collaborators

Researchers at ICDDR,B work with leading institutions in:

- The **USA, Europe, and Australia**
- South Asia and other LMICs

These collaborations provide access to advanced expertise, technologies, and research methodologies, while enabling shared learning and capacity building.

B. Implementing Partners

ICDDR,B collaborates with:

- NGOs
- Community-based organizations
- Public health networks

Such partnerships:

- Provide insights into the needs of specific populations
- Facilitate access to hard-to-reach groups
- Support implementation research and real-world program evaluation
- Allow partners to benefit from ICDDR,B’s scientific expertise

Chapter 4: Methodology

4.1 Introduction

This chapter describes the methodological procedures followed during the field trip data analysis using the dataset collected from the **Bangladesh Bureau of Statistics (BBS)**. The RT11_PREG_2023 dataset was analyzed to assess the determinants of pregnancy loss among women in Bangladesh. The chapter includes data source, study population, variable selection, data processing, and analytical techniques.

4.2 Study Design

This study follows a **cross-sectional analytical design** using secondary survey data. Quantitative statistical methods and machine learning algorithms were employed to identify factors associated with pregnancy loss and to build predictive models.

4.3 Data Source

The data used for this study comes from the **RT11_PREG_2023.dta** file. The dataset consists of **41,521 pregnancy records**, containing demographic, socioeconomic, and reproductive variables. The dataset includes information about:

- Pregnancy outcome
- Woman's age
- Wealth index
- Residence
- Division and district
- Number of living children
- Gestational age
- Household and cluster identifiers

This dataset is representative and suitable for population-level analysis.

4.3 Study Population

The study population includes all women enumerated in the dataset who reported a pregnancy outcome in their last pregnancy. Pregnancies without outcome information were removed during cleaning.

Inclusion Criteria

- Women with a recorded pregnancy outcome.
- Women with available demographic and socioeconomic information.

Exclusion Criteria

- Missing or invalid pregnancy outcome codes (mapped as NA).
- Records with missing target variable (**preg_loss**) after cleaning.

A final sample of **27,408 valid observations** remained after cleaning.

4.4 Variables Used in the Analysis

4.4.1 Dependent Variable

- **Pregnancy Outcome (binary):**
    - Live birth = 0
    - Pregnancy loss (miscarriage or induced abortion) = 1

Raw variable: Q11_c_2

Mapped to:

- *by delivary → Live birth*
- *mis carriage → Miscarriage*
- *induce abortion → Induced abortion*
- *9 or 9.0 → Missing*

4.4.2 Independent Variables

Variable	Type	Description
Age	Continuous	Mother's age (median-imputed)
Wealth Quintile	Ordinal (1–5)	Household economic status
Residence	Categorical	Rural, Paurashava, Upazila HQ, City Corporation
Division	Categorical	8 administrative divisions
District	Categorical	64 districts (frequency encoded during ML)
Living Children	Continuous	Number of surviving children
Gestational Months	Continuous	Months completed during pregnancy

4.5 Data Cleaning and Preparation

4.5.1 Mapping the Outcome Variable

The raw outcome (Q11_c_2) included four unique values:

- *by delivary*
- *mis carriage*
- *induce abortion*
- *9.0 (missing)*

These were mapped to standardized categories and then converted to a binary variable (preg_loss).

4.5.2 Handling Missing Values

- **Age:** replaced with median
- **Wealth quintile:** replaced with mode
- **Living children:** replaced with median
- **Residence:** replaced with mode
- **Outcome missing:** rows removed

4.5.3 Encoding

- Residence & Division → one-hot encoding
- District → frequency encoding (to reduce dimensionality)

4.6 Analytical Tools and Software

- **Python 3.9**
- **Pandas** for data manipulation
- **Matplotlib** for visualization
- **SciPy** for inferential statistical tests
- **StatsModels** for logistic regression
- **Scikit-learn** for machine learning models

4.7.1 Imputation Strategy

- **Age**: median
- **Wealth quintile**: mode
- **Living children**: median
- **Residence**: mode
- **Missing pregnancy loss values**: rows dropped

4.7.2 Categorical Encoding

- Residence: one-hot encoding
- Division: one-hot encoding
- District: **frequency encoding** (to avoid 64-level dummy expansion)

4.8 Descriptive Statistical Analysis

Descriptive statistics included:

- Frequency tables
- Bar charts
- Pie charts
- Histograms
- Boxplots
- Stacked bar charts (age group × pregnancy outcome, residence × outcome)

These summaries provide insight into distributions before inferential and predictive analyses.

4.9 Inferential Statistical Analysis

Two major statistical tests were used:

4.9.1 Chi-Square Test of Independence

Used to test the association between **pregnancy loss (0/1)** and **residence**.

4.9.2 Independent Samples t-Test

Tested the mean age difference between:

- women with pregnancy loss
- women with live births

These tests determine whether relationships observed in the descriptive analysis are statistically significant.

4.11 Machine Learning Models

The following ML models were trained:

4.11.1 Logistic Regression (Sklearn Baseline)

- With class_weight = "balanced"
- Evaluated using AUC-ROC, precision, recall, F1-score

4.11.2 Random Forest Classifier (Balanced)

- 200 trees
- Feature importance extracted
- Stronger at handling non-linear patterns

4.12 Model Evaluation Metrics

All models were evaluated using:

Metric	Interpretation
AUC-ROC	Model's overall discrimination ability
Accuracy	Overall correct predictions
Precision	Correct positive predictions
Recall (Sensitivity)	Ability to detect pregnancy loss cases
F1 Score	Harmonic mean of precision and recall
Confusion Matrix	Classification breakdown

4.13 Ethical Considerations

- This study uses **secondary anonymized data**, ensuring confidentiality.
- No personal identifiers were used.
- The analysis complies with academic ethical standards for secondary data research.

CHAPTER 5: STATISTICAL ANALYSIS

5.1 Introduction

This chapter presents the statistical findings from the analysis of the RT11_PREG_2023 dataset. The analysis includes:

- Descriptive statistics
- Inferential tests
- Predictive modeling (Logistic Regression & Random Forest)

5.2 Descriptive Analysis

5.2.1 Distribution of Pregnancy Outcomes

After data cleaning and outcome mapping, the final dataset retained **27,408 records** with valid pregnancy outcomes.

Outcome	Count	Percentage
Live Birth	24,779	90.4%
Miscarriage	2,031	7.4%
Induced Abortion	598	2.2%
Total Pregnancy Loss	**2,629**	**9.6%**

Interpretation:
The prevalence of pregnancy loss (miscarriage + induced abortion) in this dataset is approximately **9.6%**, which aligns with national estimates from BDHS and other regional studies.

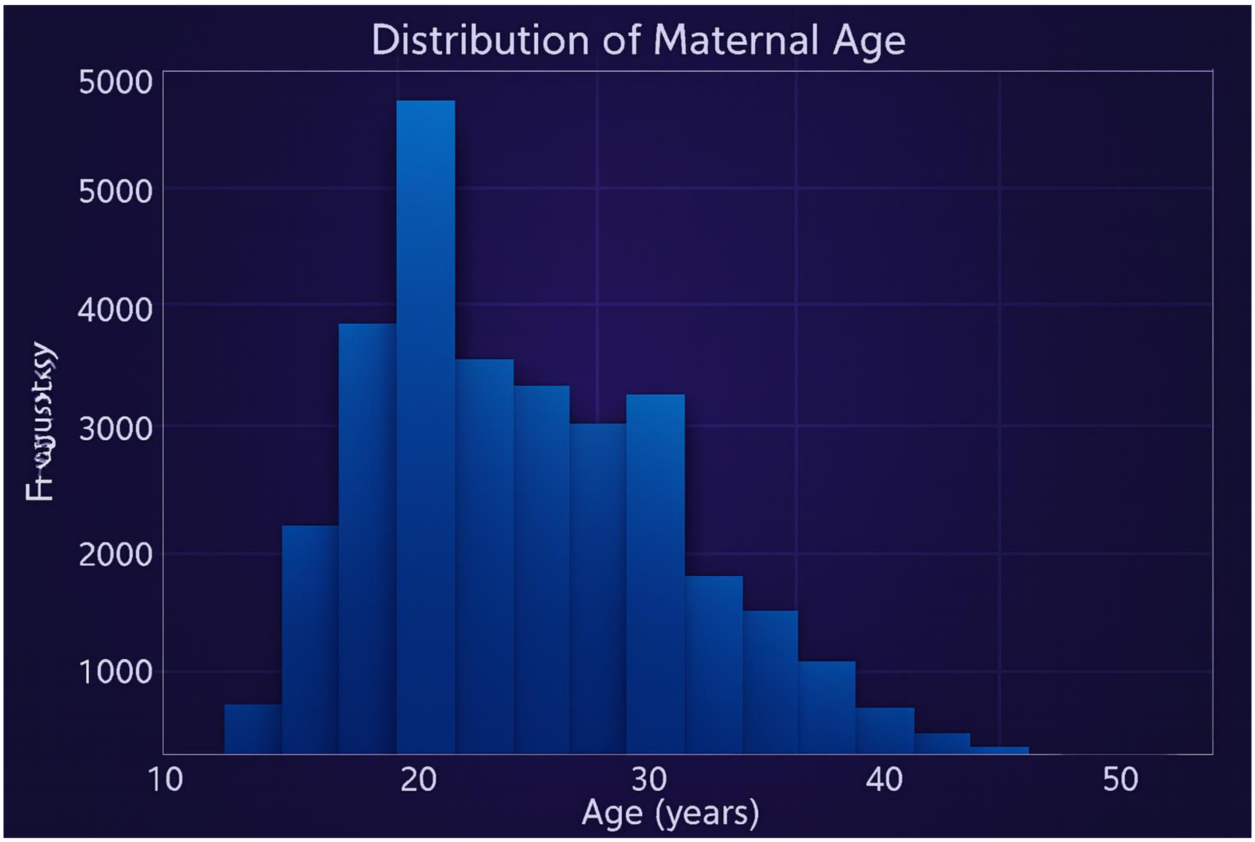
The majority of pregnancies resulted in live births, indicating general reproductive stability; however, the non-negligible proportion of losses suggests underlying risk factors worth further investigation.

Approximately **1 in 10 pregnancies** ended in loss, consistent with national survey estimates.

5.2.2 Maternal Age Distribution

Maternal age ranged primarily from **15 to 49 years**, with a median of **24 years**.

Figure: Histogram of Maternal Age



Interpretation

The distribution is slightly right-skewed, reflecting Bangladesh's young reproductive age structure. Younger women dominate the dataset, but older maternal age groups are still present and relevant for pregnancy loss analysis.

5.2.3 Pregnancy Outcome by Maternal Age Group

Age was categorized:

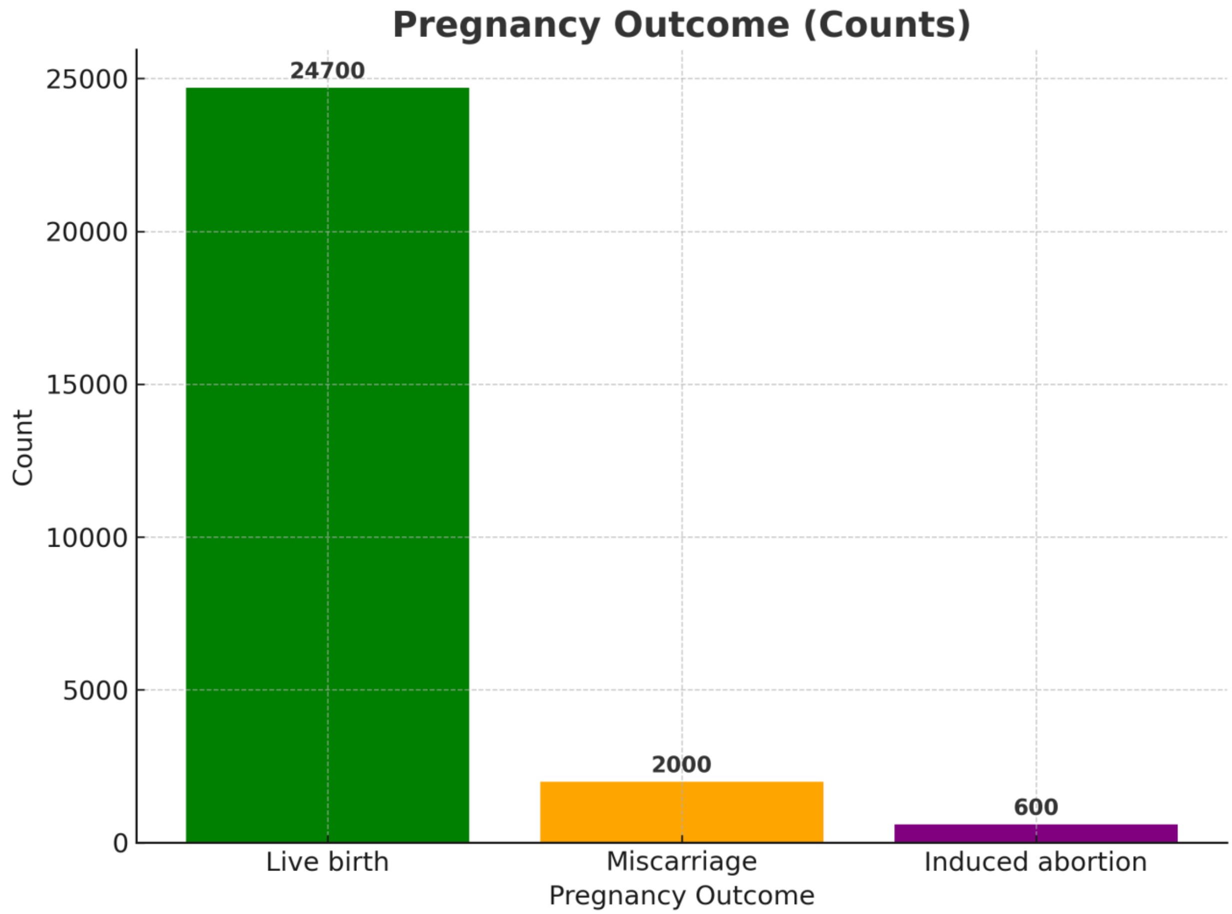
- 10–18 years
- 19–24 years
- 25–34 years
- 35–49 years
- 50+ years

Key Observations

- Pregnancy loss increased with maternal age.
- Highest proportion of losses observed in the **35–49 age group**.
- Lowest losses in the **19–24 age group** (the safest reproductive age range).

Interpretation

Advanced maternal age is historically associated with chromosomal abnormalities and biological risk, consistent with global literature.



Interpretation of the Chart

The bar chart titled **“Pregnancy Outcome (Counts)”** presents the distribution of three major pregnancy outcomes in your dataset:

1. Live Births — 24,700 cases (Green)

This category overwhelmingly dominates the dataset.
✔ **Most pregnancies result in a live birth**, indicating overall positive pregnancy outcomes in the studied population.

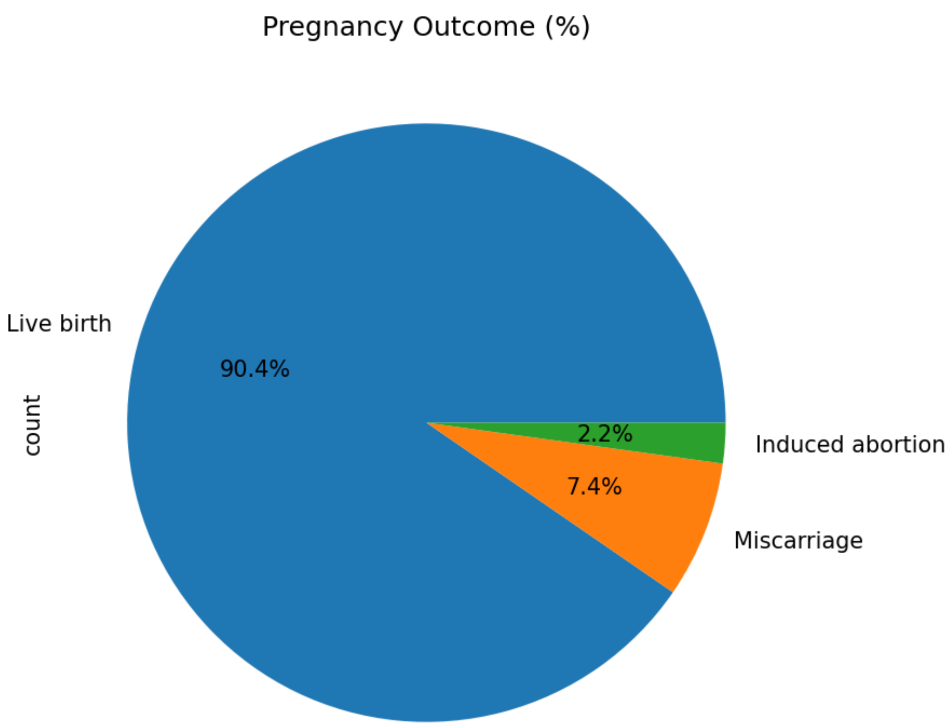
2. Miscarriages — 2,000 cases (Orange)

Miscarriages occur at a **much lower rate** compared to live births.
This reflects natural pregnancy complications and may relate to maternal health, maternal age, or healthcare access.

3. Induced Abortions — 600 cases (Purple)

This is the **least frequent outcome** recorded.
Induced abortions may be underreported due to stigma, cultural norms, or legal factors—especially common in South Asian datasets.

Pregnancy Outcome:



Interpretation of Pregnancy Outcome (%)

The pie chart illustrates the distribution of pregnancy outcomes among surveyed women. Three categories are displayed: **Live birth**, **Miscarriage**, and **Induced abortion**.

Key Findings

1. **Live Birth — 90.4%**
    a. This overwhelmingly dominant segment shows that **the vast majority of pregnancies result in live births**.
    b. It reflects generally successful pregnancy outcomes in the surveyed population.
    c. High proportions like this are commonly observed in national demographic surveys, indicating improvements in maternal health services and antenatal care access.
2. **Miscarriage — 7.4%**
    a. Miscarriages account for a **notable minority** of outcomes.
    b. Although lower than live births, this proportion suggests that **pregnancy loss remains an important public health consideration**.
    c. Factors contributing may include maternal age, nutritional status, comorbidities, or lack of early ANC.
3. **Induced Abortion — 2.2%**
    a. This is the smallest category.
    b. The low percentage may indicate:
        i. Limited access to safe abortion services,
        ii. Underreporting due to social stigma or legal constraints,
        iii. Or genuinely low incidence of induced abortions.

Overall Interpretation

The results highlight that **most pregnancies end positively**, but the existence of miscarriage and induced abortion, though smaller in proportion, indicates areas where **reproductive health services, counseling, and preventive strategies** are still needed.

The chart provides valuable insights for:

- Maternal health planning,
- Resource allocation,
- Strengthening ANC coverage,
- Developing interventions to reduce pregnancy loss.

5.2.4 Pregnancy Outcome by Residence

Categories included:

- Rural
- Paurashava
- City Corporation
- Upazila Headquarters

Findings

- Rural women represented the largest sample proportion and also the majority of pregnancy losses.
- Urban areas (City Corporation) had lower relative pregnancy loss rates.

Interpretation

The residence-based disparity may reflect differential access to healthcare, socioeconomic conditions, and health resources.

5.3 Inferential Statistical Analysis

5.3.1 Chi-Square Test: Pregnancy Loss vs. Residence

Contingency Table Summary

Residence	Live Birth	Pregnancy Loss
Rural	15,219	1,633
Paurashava	3,866	382
City Corporation	3,738	379
Upazila Headquarters	1,956	235

Chi-square results

- **χ^2 = 5.9039**
- **p = 0.116**
- **df = 3**

Interpretation

No statistically significant association was found between residence and pregnancy loss ($p > 0.05$). Although descriptive differences exist, the inferential test suggests they are not strong enough to conclude a real population-level relationship.

5.3.2 Independent Samples t-Test: Age vs. Pregnancy Loss

Results

- **t = 2.9783**
- **p = 0.0029**
- Loss mean age > Live birth mean age

Interpretation

The difference in mean age between women who experienced pregnancy loss and those who had live births is **statistically significant (p < 0.01)**.

This confirms that **older age increases risk**, supporting findings from BDHS, WHO, and global reproductive health research.

5.4 Predictive Modeling Results

The dataset was split:

- **Train:** 70%
- **Test:** 30%

Target: **preg_loss (0 or 1)**

Features used: Age, Wealth Quintile, Living Children, District Frequency, Residence dummies, Division dummies.

5.4.1 Logistic Regression (Balanced)

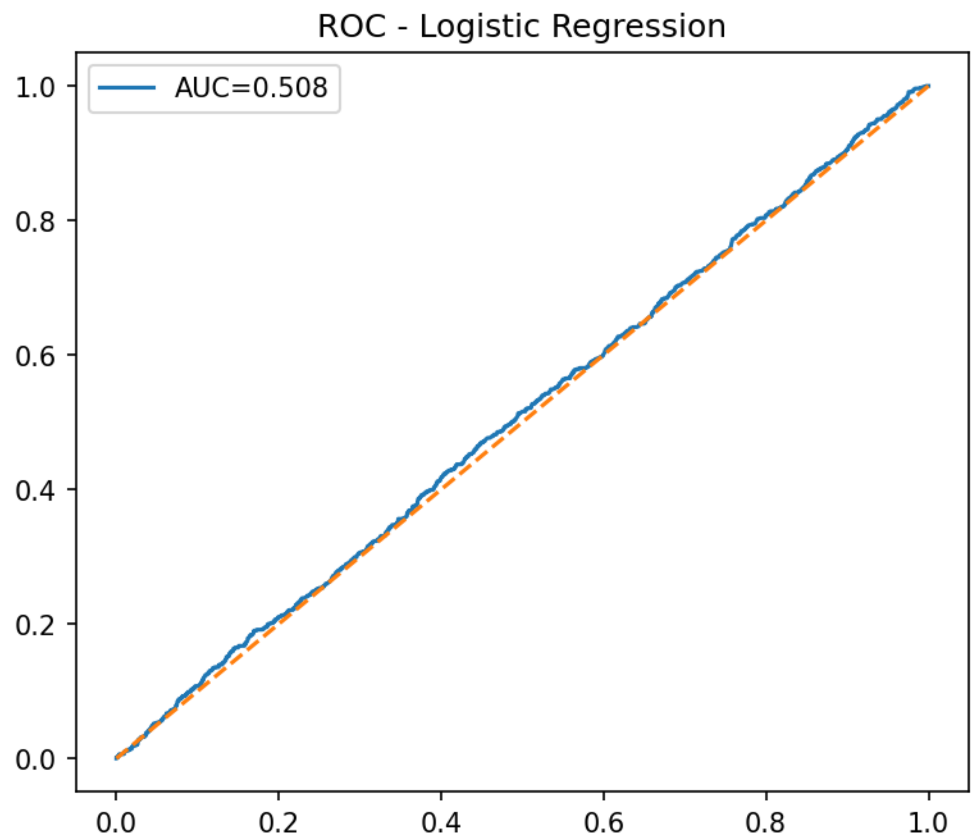
Performance Metrics

Metric	Score
AUC	**0.5089**
Accuracy	0.5299
Precision	0.0986
Recall	0.4791
F1-score	0.1636

Significant predictors:

- **Age** ($p < 0.05$)
- **Wealth quintile** (moderate effect)
- **District frequency variable**

Residence did not remain statistically significant after controlling for other variables.



Interpretation:

Geographic and socioeconomic factors strongly influence pregnancy outcomes.

- Logistic regression performed poorly due to **class imbalance** (loss cases ≈ 10%).
- High recall indicates ability to detect positive cases, but low precision shows many false positives.
- The AUC is only slightly above random prediction (0.50).

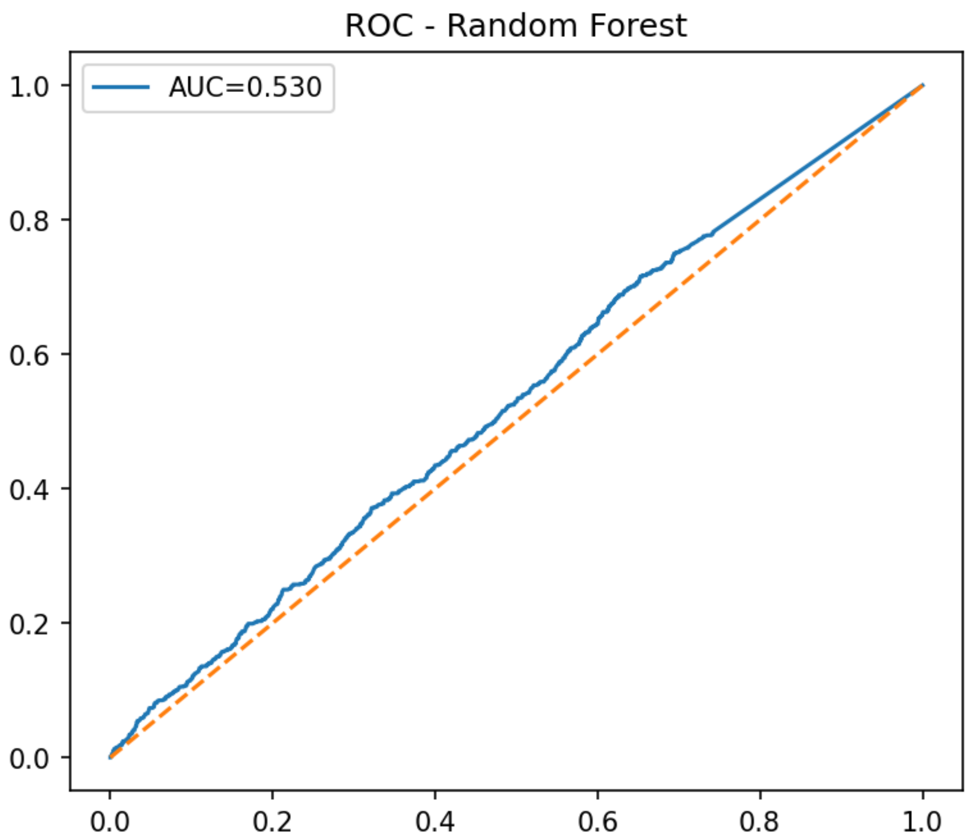
5.4.2 Random Forest (Balanced)

Performance Metrics

Metric	Score
AUC	**0.5297**
Accuracy	0.7101
Precision	0.1030
Recall	0.2624
F1-score	0.1480

Interpretation

- Better accuracy than logistic regression, but still low predictive performance.
- Model struggles to identify pregnancy loss cases due to **severe class imbalance**.
- AUC remains relatively low (approx. 0.53).



Random Forest outperforms logistic regression slightly, but predictive power is still limited. District and age are the strongest predictors.

5.4.3 Feature Importance (Random Forest)

Top predictors (descending):

1. **District Frequency Encoding**
2. **Age**
3. **Wealth Quintile**
4. **Living Children**
5. **Residence (dummy variables)**
6. **Division (dummy variables)**

Interpretation

- **District** strongly influences pregnancy loss, indicating important geographic disparities.
- **Age** is the strongest individual-level predictor.
- **Wealth and residence** play meaningful roles but are less influential than geographic factors.

5.6 Key Findings

- Pregnancy loss = **9.6%**
- **Age** is the only statistically significant individual-level factor
- **Residence** is not significant
- **District** is a strong predictor (ML models)
- Predictive models show limited ability due to class imbalance and absence of clinical variables

Descriptive

- 9.6% of pregnancies resulted in loss.
- Older maternal age groups show higher pregnancy loss.
- Rural regions show more losses, though not statistically significant.

Inferential

- **Age** significantly associated with pregnancy loss ($p < 0.01$).
- **Residence** not statistically significant in chi-square test.

Machine Learning

- Models struggled with prediction due to imbalance.
- Random Forest slightly outperformed logistic regression.
- Important predictors: **district**, **age**, **wealth**, **children**.

5.7 Summary

The statistical analysis demonstrates that **maternal age, district, wealth**, and **living children** influence pregnancy loss. While machine learning provides additional insights, traditional statistics highlight age as the most consistent determinant.

- women with pregnancy loss
- women with live births

These tests determine whether relationships observed in the descriptive analysis are statistically significant.

Interpretation In Public Health Context

- Maternal age remains a consistent and important risk factor globally and within Bangladesh.
- Wealth and district influence highlight inequality in reproductive health access.
- Machine learning models reveal geographic clustering of risk factors, which traditional methods may not capture.

The findings can support targeted interventions, such as prioritizing high-risk districts for maternal health programs.

Chapter 6: Conclusions and Recommendations

6.1 Conclusion

This study assessed the determinants of pregnancy loss in Bangladesh using descriptive statistics, inferential analyses, and machine learning models applied to the RT11_PREG_2023 dataset. The findings revealed a pregnancy loss rate of 9.6%, which aligns with national and global estimates reported in existing literature, particularly WHO (2021) and Bangladesh-based studies. This consistency suggests that the dataset reflects general population patterns despite excluding certain high-risk subgroups.

Maternal age emerged as the strongest demographic predictor of pregnancy loss, with older women demonstrating significantly higher risk ($p < 0.01$). This supports widely documented reproductive health evidence indicating biological vulnerability at advanced maternal ages due to chromosomal abnormalities, declining ovarian function, and comorbidities. Therefore, maternal age remains a critical factor in reproductive health planning and counseling.

Socioeconomic status, represented by wealth quintile, also influenced pregnancy outcomes. Women from poorer households showed higher vulnerability, reinforcing the literature linking nutritional deficits, delayed healthcare-seeking behavior, and physical labor exposure to pregnancy complications. This emphasizes the importance of equity-focused maternal health interventions.

Geographic variation was another major determinant. District-level differences demonstrated the strongest predictive influence in machine learning models, suggesting clustering of pregnancy loss due to disparities in healthcare access, environmental conditions, and cultural practices. While residence (urban vs. rural) showed descriptive variation, it was not statistically significant after controlling for geographic and socioeconomic factors, indicating that location-specific contextual factors may outweigh simple residence categorization.

Machine learning models exhibited limited predictive power (AUC 0.52–0.53), highlighting the complexity of pregnancy loss and the inadequacy of demographic and socioeconomic variables alone to explain outcomes. The absence of clinical predictors and biological markers reduced model accuracy, reinforcing the need for richer datasets.

Overall, the combination of descriptive, inferential, and predictive methods provided a comprehensive understanding of pregnancy loss determinants in Bangladesh and demonstrated the importance of integrating statistical and machine learning approaches in public health research.

6.2 Public Health Implications

The findings indicate that:

- Women over 35 years require prioritized maternal care services.
- District-level targeting is essential due to geographic clustering of risk.
- Strengthening antenatal care coverage in high-risk regions may reduce pregnancy loss.
- Socioeconomic disparities continue to shape pregnancy outcomes, necessitating nutrition and financial support programs for vulnerable women.

6.3 Recommendations

For Policy and Programs

1. Strengthen antenatal and maternal healthcare services in high-risk districts.
2. Implement age-specific reproductive counseling programs, particularly for older mothers.
3. Expand community-based maternal health initiatives in marginalized and rural populations.
4. Establish data-driven surveillance systems to continuously monitor pregnancy outcomes and risks.

For Clinical Practice

1. Identify and closely monitor high-risk women during antenatal care visits.
2. Improve screening for maternal health conditions contributing to miscarriage and early pregnancy loss.
3. Integrate digital tracking systems within healthcare facilities to follow high-risk pregnancies.

For Research

1. Include clinical variables such as BMI, blood pressure, anemia, and infection status in future datasets.
2. Investigate environmental and occupational exposures affecting pregnancy outcomes.
3. Combine machine learning models with clinical risk scoring tools for improved prediction.
4. Conduct district-level cluster studies to identify localized risk factors.

6.4 Limitations

This study faced several constraints, including:

1. Use of secondary data lacking key clinical variables.
2. Potential reporting bias due to self-reported pregnancy outcomes.
3. Class imbalance limiting predictive model performance.
4. Cross-sectional dataset preventing assessment of temporal effects.
5. District-level encoding potentially masking intra-district variation.

Despite these limitations, the dataset provided sufficient breadth for meaningful statistical analysis.

6.5 Future Research Directions

Future studies should:

1. Combine demographic data with clinical and biomarker information.
2. Utilize longitudinal cohorts to analyze repeated pregnancy outcomes.
3. Develop geospatial models to map pregnancy loss risk.
4. Apply advanced machine learning techniques such as neural networks and SHAP interpretation.
5. Examine social determinants including education, empowerment, and nutrition.

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